



GLAB 927.1.2:

Hands-On Text Processing for Email Management with Python

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Introduction

In this lab, you will learn how to use Python's NLTK library for key text processing techniques such as stemming and lemmatization, applied to the context of managing and categorizing business emails. These techniques will help you streamline and analyze large volumes of email data more effectively.

Objectives

By the end of this lab, learners will be able to:

- Apply concepts of stemming and lemmatization in the context of email management.
- Implement stemming using Python's NLTK library to process and categorize email content.
- Apply lemmatization using Python's NLTK library to standardize email content for improved analysis.

Equipment

Access to a Python environment with NLTK installed.

Submission

Submit your Python file showcasing the implementation of stemming and lemmatization for email management scenarios using the NLTK library. Click the "Start Assignment" button in the top-right corner of the assignment page in Canvas to upload your file. Your submission should demonstrate how these text processing techniques can be effectively applied to business email data.

Instructions

If you do not have a Python environment ready, you can use the [link](#) to install python:

Task 1: Importing Libraries

TO-DOs - 1:

Begin by importing the necessary libraries for this hands-on exercise. Ensure NLTK is installed in your Python environment.

If NLTK is not installed, use the following command to install it in your terminal:

```
pip install nltk
```

```
# Importing NLTK library

import nltk

# Checking NLTK version

nltk_version = nltk.__version__

# Print confirmation message and NLTK version

print(f"NLTK library successfully imported. NLTK version: {nltk_version}")
```

Task 2: Implementing Stemming

In this task, imagine that you are managing a large volume of business emails. Stemming helps reduce words to their root forms, making it easier to categorize similar emails or extract common themes from email content.

TO-DOs - 2:

Apply stemming to a sample email using NLTK. This will simplify the email text by reducing words to their base forms, which aids in categorizing and analyzing email content.

```
from nltk.stem import PorterStemmer

# Download NLTK resources (tokenizers)

nltk.download('punkt')

# Sample email content for stemming

email_content = "Our team will be working on the upcoming project deadlines. Please ensure that all deliverables are submitted before the end of the week."

# Create a Porter Stemmer instance

porter_stemmer = PorterStemmer()

# Apply stemming to each word in the email content

stemmed_words = [porter_stemmer.stem(word) for word in nltk.word_tokenize(email_content)]

# Print the original and stemmed text

print("Original Email Content:", email_content)

print("Stemmed Email Content:", " ".join(stemmed_words))
```

Task 3: Applying Lemmatization

In this task, you will use lemmatization to standardize the text from emails. Lemmatization converts words to their base or dictionary form, which helps in creating consistent and analyzable data from varied email content.

TO-DOs - 3:

Apply lemmatization to a different sample email using NLTK. This process will standardize the email text, making it easier to analyze and understand the content.

```
from nltk.stem import WordNetLemmatizer

# Sample email content for lemmatization

email_content_lemma = "The team members are working on their tasks diligently. Please ensure
that every report is submitted promptly."

nltk.download('wordnet')

# Create a WordNet Lemmatizer instance

lemmatizer = WordNetLemmatizer()

# Apply lemmatization to each word in the email content

lemmatized_words = [lemmatizer.lemmatize(word) for word in
nltk.word_tokenize(email_content_lemma)]

# Print the original and lemmatized text

print("Original Email Content:", email_content_lemma)

print("Lemmatized Email Content:", " ".join(lemmatized_words))
```

Congratulations! You have successfully explored and implemented text processing techniques for email management using NLTK. Your skills in stemming and lemmatization will enhance your ability to handle and analyze email content effectively.